AgriPreneurs

The Buildathon

Build for the Farmer

Kickoff Keynote

Sunday, 4 October 2026 5th Floor, RTIH, VMRDA Deck, Siripuram, Visakhapatnam

An initiative of AgriPreneurs · Partners: AI for Vizag, AI Karyashala, Quantamizers, GDG Vizag · Support: RTIH, G-TEC, VDC

01

Why we are here

The farmer grows the food we all eat, and keeps the least of it.

A farmer carries the loss alone

- **Rot before sale** On many Vizag farms, 30 to 40% of what is grown spoils before it reaches a buyer.
- **The margin leaks out** A long chain of middlemen takes the margin. The farmer keeps the least.
- **Advice that sells** Guidance comes from the person selling the input, so more chemical goes in than is needed.
- **Good ideas die** Local solutions never get carried from a field into a business.



The farmer comes first, and the rest follows.

02

The context

Why startups built for the farmer need a Buildathon.

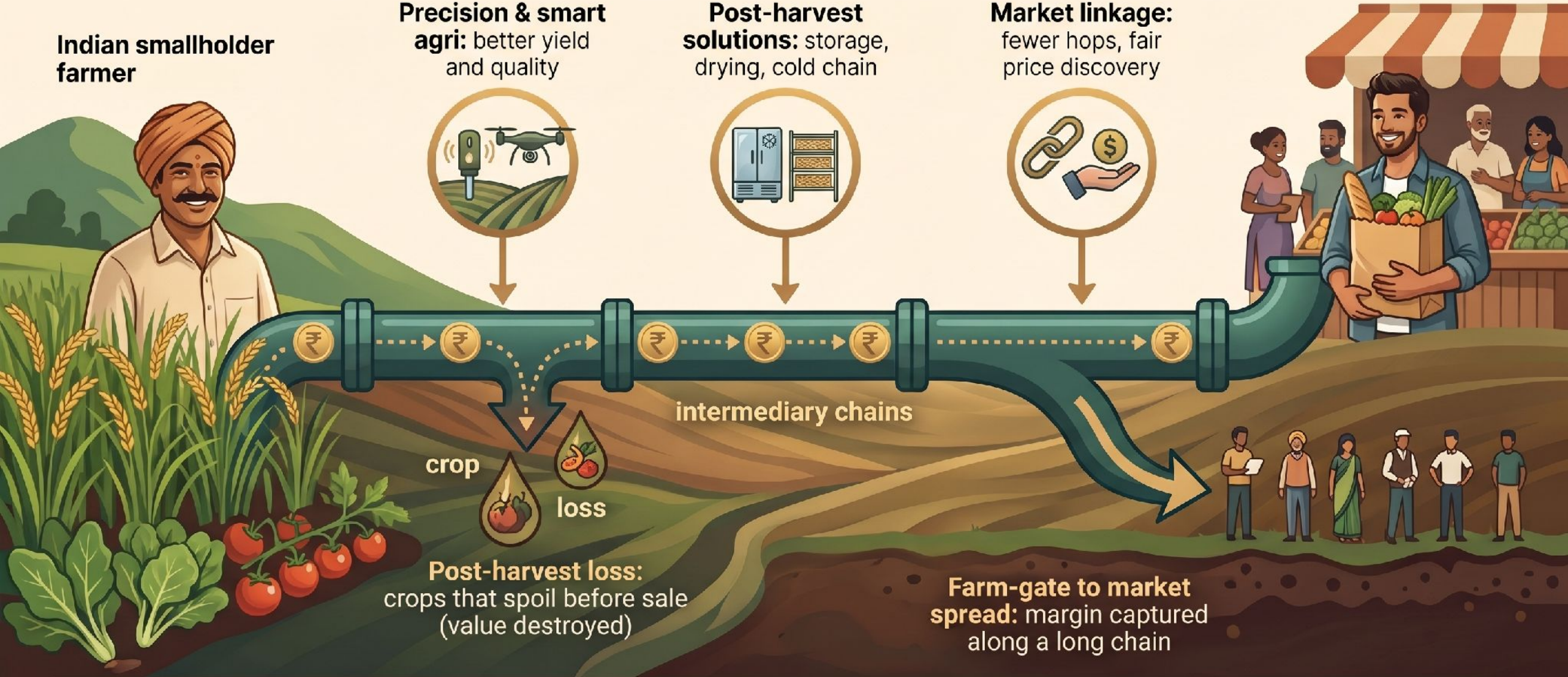
Where the value Leaks, and how the Farmer wins

Indian smallholder farmer

Precision & smart agri: better yield and quality

Post-harvest solutions: storage, drying, cold chain

Market linkage: fewer hops, fair price discovery



Post-harvest loss: crops that spoil before sale (value destroyed)

Farm-gate to market spread: margin captured along a long chain

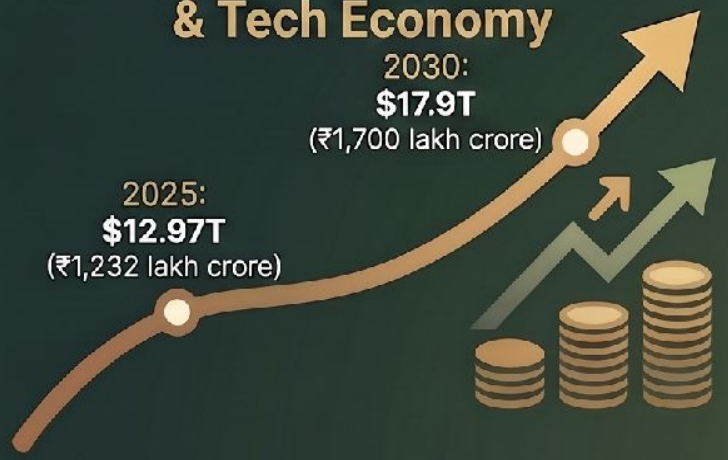
The customer pays about the same. The farmer's share rises. What shrinks is waste and middle-margin, not the shopper.

Illustrative. India agritech value pools, Inc42-StarAgri 2025; market linkage ~45% of value by 2030.

The Global Macro-Context: Why Now?

GLOBAL

The Agri, Food Processing & Tech Economy



The driver: Agri 4.0



Smart & precision-agri tech



2025: **~$32B** (~₹3.04 lakh crore) (Approx.)
2030: **~$65B** (~₹6.18 lakh crore) (Approx.)

INDIA

Agri 4.0

India's Agri, Food Processing & Tech Economy



2025:
~₹57 lakh crore
(~$600B)

2030:
~₹85 lakh crore
(~$900B)

India agritech market



2030: **$28B**
(₹2.66 lakh crore)
~25% CAGR

The headroom



Agritech is under **2%** of **India's Agri, Food Processing & Tech Economy** today, only **~5% by 2030**

Sources: Market Intelligence; Industry Reports; Indian Government Data (IBEF/NSO). **All figures are illustrative approximate projections. USD-INR ₹95 rate, Aug 2026 forecast.

Start in the Field, Not on the Whiteboard

1. Get to know them



2. See the real issues



3. Create simple solution



4. Put it to the test



5. Fix the losses



6. Connect to markets



With Nature, With Farmer.

03

The framework

How an idea becomes a farmer-first startup, and where you fit.

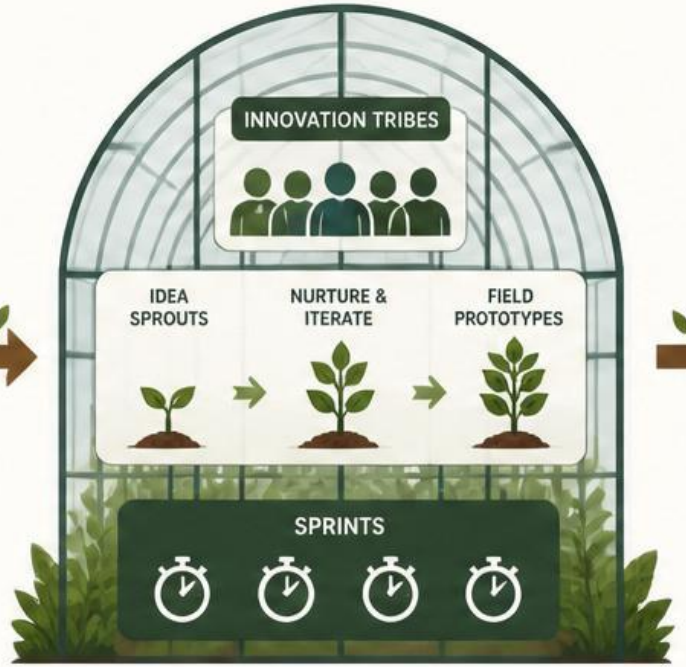
The Seed Bank

Collecting diverse ideas and planting potential.



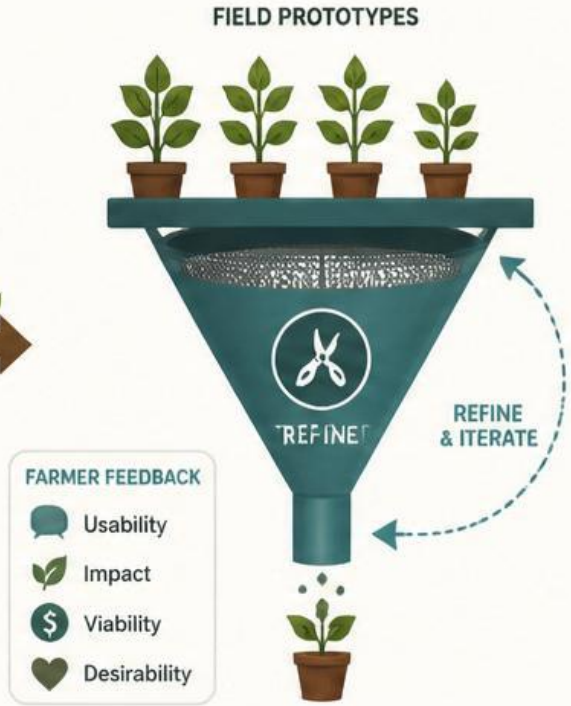
The Greenhouse

Nurturing ideas through sprints and collaboration.



The Pruning

Testing, refining, and filtering for strongest potential.



The Harvest

Investable startups ready for market impact.



Diverse ideas from many sources are collected, curated, and planted in the Hive.

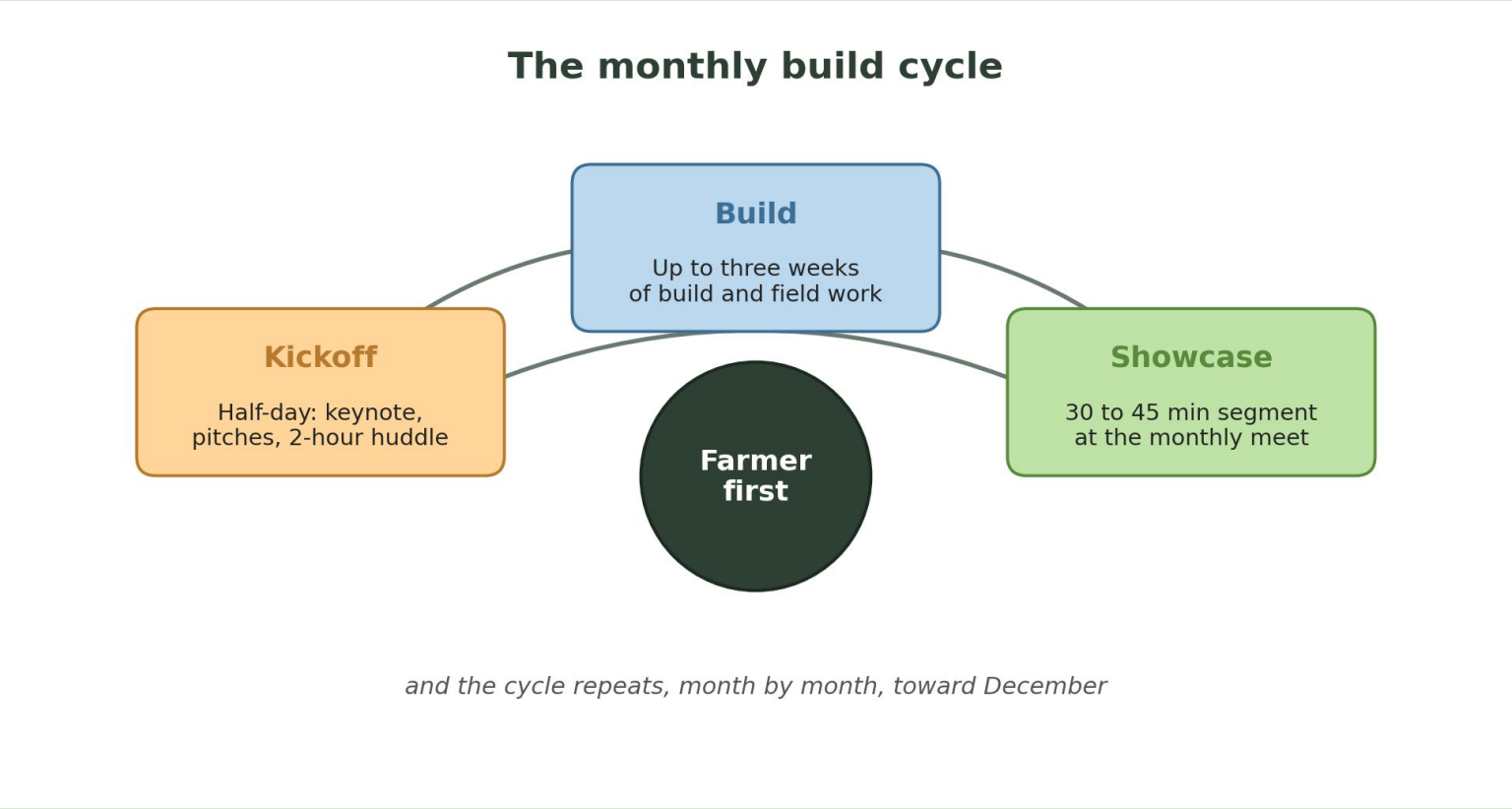
Cross-functional tribes nurture ideas through time-boxed sprints, turning sprouts into field-ready prototypes.

Prototypes are tested with real farmers. Feedback prunes weak ideas and strengthens the best.

Top-performing startups are harvest-ready—validated, investable, and positioned for market impact.

From seed to impact.
Growing innovation. Cultivating change.

The Buildathon cycle: a monthly rhythm



1. **Kickoff**
   Half day. Keynote, ideas, a two-hour huddle to join a Tribe.

2. **Build**
   Up to three weeks. Build and test in a real village.

3. **Showcase**
   A segment at the monthly meet, with real farmer numbers.

4. **Repeat**
   Month by month, toward the Mela in December.

What can each of you build

Who you are	What you bring	What you build
Student	Energy, curiosity, time	A first prototype, field data, a demo a farmer can hold
Developer	Python, Java, React, Node, AI	The farmer-facing tool: price signal, grading, advisory, listing
Designer	Make it simple for a farmer	Plain Telugu flows, screens a first-time user understands
Product / BA	Frame the problem, see the model	A sharp problem, the field metric, the path to first revenue
IoT / hardware	Sensors, boards, tinkering	Monitoring for storage, simple field sensing
Mentor / farmer	Ground truth and judgement	Guidance, validation, a real farmer to test with

You build. We walk with you. The farmer wins.

You build

- One small thing, not a platform
- A real, farmer-validated problem
- Build with tools you know, or vibe-code it
- Something a farmer actually uses

We provide

- The problem, picked with you
- Help thinking it through, and building
- Real farmers to test with, in the field
- If it works, funding and next steps

The farmer wins

- Less waste, more of the crop sold
- A fairer price, paid faster
- Honest advice, not a sales pitch
- A tool made with them, not for them

Problems worth your weekend

Organic marketplace

Produce, and value-added lines like vegetable powders and dehydrated veg, to a real buyer. On ONDC and the Digi Rythu Bazaar.

Daily price signal

A Telugu WhatsApp reply with today's mandi price, from open government data.

Grade-assist

A photo or checklist that grades a lot, so quality gets paid for.

Crop-photo check

Send a leaf photo, get a likely issue and honest, unbranded advice.

Storage booking

Book a slot, hold a receipt, sell later instead of at the glut.

Urban automated farming

City and rooftop growing run by small robots and sensors, a farm closer to the buyer.

Plus mud storage, a weed mat, hyperspectral seed analysis, a cold-chain tracker, and your own idea.

Rather build an app? Start here.

Not everyone works on core modules. Build the applications a startup needs, on top of them.

Organic marketplace app

Lists produce, takes orders, issues receipts

builds on M1 · M6 · M8 · M9

Telugu advisory app

Ask, get a plain answer and a sowing calendar

builds on M1 · M2 · M5

Price-alert WhatsApp bot

Text a crop, get today's nearest price

builds on M1 · M2 · M4

Venture dashboard

What a venture did, reached, and earned

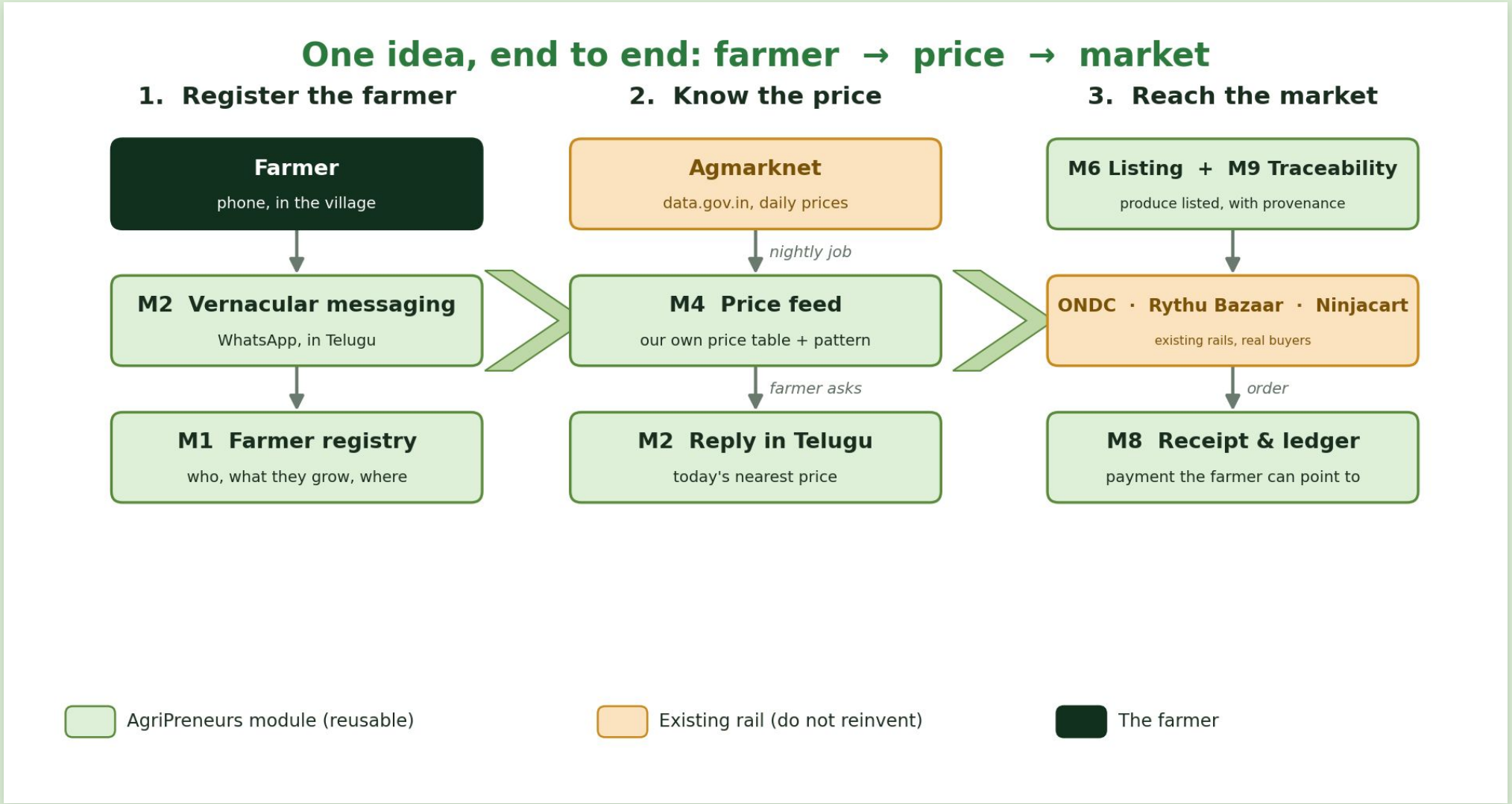
builds on M1 · M6 · M8

Mandi price tracker

A daily price pattern for a few crops

builds on M4

Design and architecture of one idea



Three small steps, each reusing a module and standing on a rail that already exists. Build the thin layer, not the whole stack.

What to build, and what we provide

A Tribe builds, by 16 October

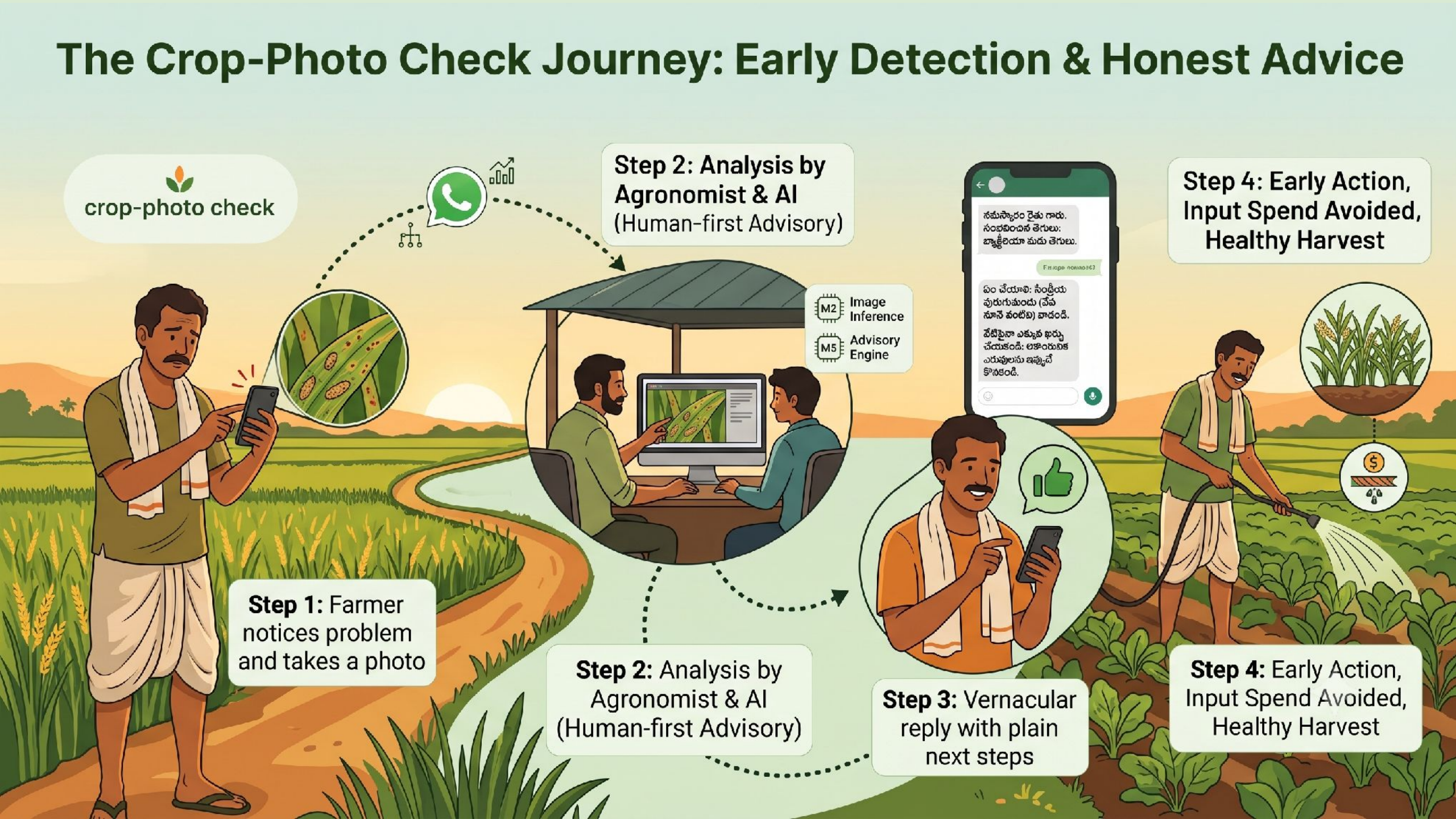
- Registry and WhatsApp hook (M1, M2)
- Nightly price pull and a Telugu reply (M4)
- One listing on a real rail (M6), QR tag (M9)
- A simple receipt when an order lands (M8)
- Onboard five real farmers, land real orders

We provide, from here

- Farmer and FPO contacts, and the venue
- An intro to ONDC and the Digi Rythu Bazaar
- An Agmarknet starter and the open modules
- Mentors, and vibe-coding help
- A path to funding if it works

From here, today: form your Tribe, pick the one slice you can ship, commit to five farmers by 16 October.

The Crop-Photo Check Journey: Early Detection & Honest Advice



04

Today, and the next two weeks

What happens now, and what you do until 16 October.

Thank you for the overwhelming response.

Here is what we see in the room today.

124
builders, in person

74%
students or interns

65
bring Python

76
came to build for farmers

A build-first, Python-led crowd, concentrated in the Vizag belt, here to make something real for farmers.

Before we start: a quick read of the room

No long intros today, so hands up. This helps us place you well.

Your skill and experience

Hands up: who codes most days? First build this year? Touched hardware or IoT? Talked to a farmer about their work?

What pulled you here

One tap: to build something real, to learn, to meet people, to help a farmer, or just curious.

Participation certificate

Hands up if you would like one, so we know to prepare them.

The skills we have, and the people we need

Strong in the room

- Python, the biggest group
- Design, React, Java
- Data and a few electronics hands
- A real founder and aspiring-founder segment

We need more of

- Farmers and farm domain voices
- Product managers
- Mentors who have built and shipped

The ask: bring someone you know who is a farmer, a product person, or a mentor. Every Tribe needs one.

Today's run of show

9:30	Welcome and the Ground Truth keynote
10:15	Ideas on the table: new and existing, five minutes each
11:00	Form your Huddle: match by skill to a problem
11:30	Build huddle: lock your one job and the three-week plan
12:45	Share and commit: each Tribe names its build and its farmer number
1:00	Close

Leave with this. Build until 16 October.

You leave today with

- A real problem, picked with a farmer
- A Tribe, and your one job
- A plan: the smallest thing you can ship
- The tools to build it, vibe-coding welcome
- A farmer or field contact to test with

Until 15 October

- Build the smallest real version
- Put it in a farmer's hands, in the field
- Measure before and after, one honest number
- Note what is not built yet, and say so
- Bring it to the 17 October meet

Show your work at the AgriPreneurs meet.

On 17 October your Tribe gets the floor. A short segment at the monthly meet to show what you built, with a real farmer number. No slides for their own sake. A working thing, and what the farmer said.

The best builds graduate into Soil-Grown Ventures, **aimed at the AgriPreneurs Day @ Visakha Organic Mela in December.**

Help us reach the next builder, and the next farmer

Follow us, share the day, and tag yourself. Post your photos and short videos with #AgriPreneursBuildathon.

Website

agripreneurs.io

LinkedIn

/agripreneurs-community

Instagram

@agripreneurs.io

YouTube

@AgriPreneurs.Ecosystem

GitHub

/agripreneurs

Join the WhatsApp community: ask a volunteer for the invite link.

AgriPreneurs

Build for the Farmer.

Pick your problem. Find your Tribe. Then build.

Register · luma.com/8g3jekz7

Working with nature, not against it. From Vizag to the world.

Team AgriPreneurs · Cultivate | Connect | Create

THE PROBLEM

Farmers Sell at the Glut Because They Cannot Wait

Lowest prices tend to be lowest right after harvest (WDRA)

Traders are the main users of warehouse receipts, not farmers (NERL 2022, via a study)

₹1,000 cr guarantee corpus for receipt-backed loans (launched Dec 2024)



Cash Pressure

Loans fall due and household costs cannot wait, so farmers sell straight from the field.



No Easy Slot

Farmers rarely know which registered stores nearby have space, or what storage costs.



Receipts Hard to Use

Awareness of negotiable receipts is low and bank procedures are seen as complicated.



Price Risk

Storing pays only if prices rise, and a stored lot with no buyer can add cost and debt.

Why Current Options Fall Short

WDRA warehouses: Trusted, valid receipts, but booking is rarely simple and pledge loans cover mainly non-perishables

Pledge loans: Subsidised, but need bank visits, paperwork and eligible commodities

Local godowns: Close and easy, but give no standard receipt to borrow or sell against

THE SOLUTION, SIMPLIFIED

Storage Booking: Book, Hold, Sell Later

1 • BOOK

Reserve a Slot Before Harvest

Problem: Farmers do not know which registered stores nearby have space, or what it costs.

- List of registered warehouses and cold stores in reach
- Live space, rate per quintal per month and insurance shown
- Reserve with a small UPI advance

Goal: a slot guaranteed before the glut

2 • HOLD

A Digital Receipt That Works

Problem: Paper slips and informal godowns give nothing a bank or buyer will accept.

- Registered store issues an e-receipt to the farmer's account
- Receipt carries quantity, grade and insurance
- Optional pledge loan from a partner bank in one tap

Goal: stored produce becomes usable collateral

3 • SELL

Sell When the Price Is Right

Problem: With a receipt but no price signal or buyer, farmers still sell in a hurry.

- Daily price alert for the stored crop
- Sell on e-NAM or ONDC to verified buyers
- Loan cleared at sale; balance paid by UPI

Goal: sell by choice, not by pressure

Pilot One long-shelf-life crop such as dry chilli, one district, three registered stores, one FPO and one partner bank. Perishables need cold stores and shorter windows, so add them later. Scope is illustrative.

WORKED EXAMPLE (ILLUSTRATIVE)

Is Waiting Worth It? 20 Quintals of Dry Chilli

Sample Digital Receipt

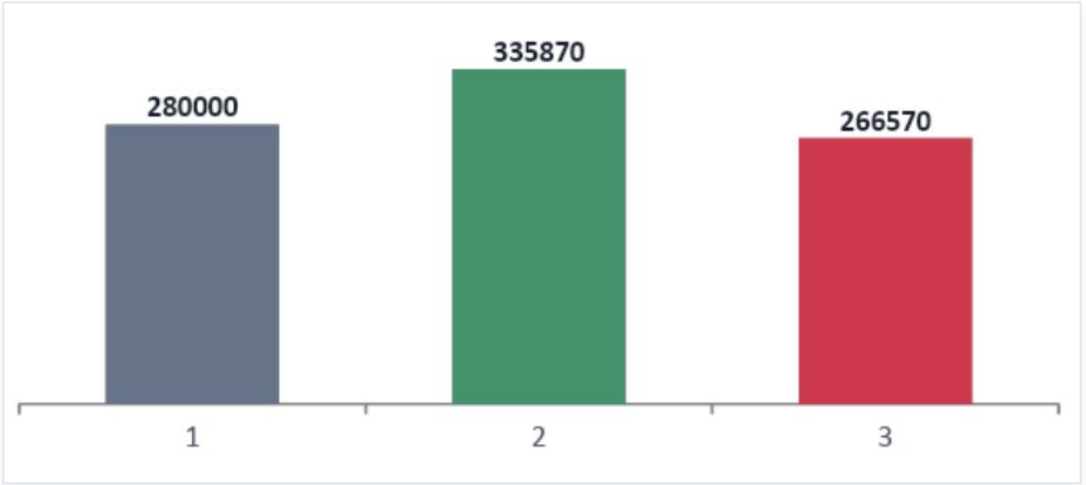
SAMPLE ONLY

Storage Receipt • Lot SR-2041

Issued by a registered warehouse

Commodity	Dry chilli, 20 quintals, Grade A
Rate	₹120 per quintal per month
Insurance	Covered while stored
Loan	Up to 70% of value, if pledged
Sale	Via e-NAM or ONDC buyers

Net Cash After 3 Months for 20 Quintals (₹)



Assumes glut price ₹14,000 per quintal, later ₹17,500. Costs: storage ₹7,200, pledge interest ₹3,430 (7%, 70% loan), 1% shrinkage.

Takeaway

Waiting pays only if the price rises by roughly ₹700 per quintal to cover costs. If flat, the farmer ends about ₹13,400 worse off, so price alerts matter.

Risks and Measures

Risks: price falls, quality loss, receipt fraud, loan stress. Measure: share of farmers selling above glut price, net gain after costs, slot fill rate, loans repaid on time.

THE PROBLEM

Farmers Sell Without Knowing Today's Price

~16,000 rows a day in the open mandi price feed, in English tables

3 prices min, max and modal per variety per market, with no single answer

10 rows per request on the public demo API key, so a full pull is slow



Language Gap

Open prices sit in English portals and developer APIs. The price sites I found are English or Hindi, not Telugu.



Rows, Not an Answer

A farmer asks "what is chilli fetching today?" but gets rows by market and variety, quoted per quintal.



Late or Thin Data

The feed fills through the day and some markets report a single trade. Stale numbers can mislead.



Trader Sets Price

Without a trusted reference, the buyer's quote at the point of sale becomes the price.

Why Current Options Fall Short

Agmarknet and data.gov.in: Authoritative, but built for portals and developers, not for a farmer's phone

Private price sites: Convenient, but English or Hindi, ad-supported, and unclear on source and date

WhatsApp alert tools: Exist for businesses, but I found no free, Telugu-first reply that cites its source

THE SOLUTION

A Telugu WhatsApp Price Signal from Open Data

1 • SOURCES

Credible, Cross-Checked Data

Problem: One feed can be late, thin or wrong for a market on a given day.

- Core: Agmarknet daily mandi data via data.gov.in
- Cross-check with e-NAM and state market data where published
- Flag stale or single-trade prices; show date and source

Goal: every number has a source and date

2 • ANSWER

One Clear Answer in Telugu

Problem: Rows of min, max and modal prices per quintal are not an answer a farmer can use.

- Farmer sends a crop name in Telugu, text or voice note
- Reply: modal price per quintal and per kg, range, 7-day trend
- Shows the nearest three markets, in local crop names

Goal: a clear price in one message

3 • REACH

Free, Simple, Trusted Reach

Problem: Farmers will not install another app or learn another portal.

- Runs on WhatsApp, which farmers already use
- FPOs and village agri centres share the number
- Optional daily alert for chosen crops after data refresh

Goal: farmers check before they sell

Build A daily job pulls the open feed after the afternoon refresh and stores its own history. A WhatsApp Business number answers queries from that store. Start with 10 crops in 5 Andhra Pradesh districts (illustrative).

HOW IT WORKS (ILLUSTRATIVE)

From a Telugu Question to a Sourced Price

Message Flow

1. **Ask:** Farmer sends a crop name in Telugu, for example chilli, by text or voice
2. **Match:** Bot maps the crop and variety and picks the nearest markets
3. **Fetch:** It reads the stored open data and cross-checks other sources
4. **Reply:** One Telugu message: price, range, trend, date and source
5. **Learn:** Optional daily alert, plus a thumbs-up or down on accuracy

Sample WhatsApp Reply

SAMPLE FIGURES

మిర్చి (Chilli) – గురజాల మార్కెట్

సగటు ధర: ₹19,000 / క్వింటాల్

అంటే సుమారు ₹190 / కిలో

కనిష్ఠం ₹14,000 • గరిష్ఠం ₹21,500

గత 7 రోజులతో పోలిస్తే: ↑ ₹500

తేదీ: 17-07-2026 • మూలం: Agmarknet

ఇది సూచన మాత్రమే. అమ్మే ముందు మీ మార్కెట్లో ధర సరిచూసుకోండి.

Takeaway

One message replaces a table of rows: the farmer sees the price per quintal and per kg, the range, trend, date and source, in Telugu.

Risks and Measures

Risks: stale or thin data, wrong variety matching, WhatsApp costs at scale, price shown without advice. Measure: accuracy against source, repeat users, farmers who check before selling.

THE PROBLEM

Quality Is Not Paid For Because It Cannot Be Shown

Optional quality assaying on e-NAM is not mandatory

By hand grain grade is often judged from a rubbed handful (Telangana officials)

Majority of APMCs lacked working assaying labs (2018 panel report)



Subjective Grading

The buyer or officer judges by eye and touch, so the same lot can get different grades.



Mixed, Dusty Lots

Mixed varieties, dust and foreign matter in field-fresh lots hide the good produce.



No Proof Travels

Buyers re-inspect each lot anyway, so a farmer's own grading carries no weight.



One Average Price

Good and poor lots fetch similar prices, so farmers lose the reason to sort and grade.

Why Current Options Fall Short

Market assaying labs: Accurate, but sit inside markets, are optional, and often under-equipped

AI sorting machines: Objective, but fixed at packhouses and procurement centres, not the farm gate

Phone AI apps: Promising, but need crop-specific training data and struggle with light and mixed lots

THE SOLUTION, SIMPLIFIED

Grade-Assist: Three Grades, Four Checks, One Card

1 • SIMPLE STANDARD

One Crop, Three Grades

Problem: Full grading standards have many parameters that a farmer cannot apply at the gate.

- Start with one visual crop, such as tomato
- Four visible checks: size, colour, damage, cleanliness
- Telugu picture guide shows Grade A, B and C

Goal: grade a lot in two minutes

2 • CHECKLIST FIRST

Tap, Snap, Done

Problem: AI-only grading needs large image sets and can fail in poor light or mixed lots.

- Farmer taps answers to the four checklist questions
- Three photos (top, side, spread tray) serve as proof
- AI only suggests and flags mismatches, added later

Goal: works on day one with no AI

3 • PROOF THAT PAYS

A Grade Card Buyers Trust

Problem: Buyers re-inspect every lot, so the farmer's grade is worth nothing at the sale.

- Grade Card (image plus QR) shared on WhatsApp or ONDC
- FPO grader spot-checks 1 in 10 lots; builds a trust score
- Buyers pre-agree grade-wise price bands

Goal: Grade A earns a Grade A price

Pilot One FPO, one crop, one buyer group. Phone only, with no new hardware. Add AI, more crops and a moisture meter for grains only after the checklist proves itself. Scope is illustrative.

HOW IT WORKS (ILLUSTRATIVE)

From Four Taps to a Grade Card

Grading Flow

1. **Pick:** Farmer chooses the crop and lot weight on a WhatsApp or app screen
2. **Tap:** Answers four questions on size, colour, damage and cleanliness
3. **Snap:** Takes three photos of the lot as proof, in plain light
4. **Card:** Gets a Grade A, B or C card with a QR code in seconds
5. **Sell:** Shares it with buyers and sells at the agreed grade price

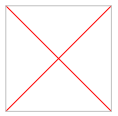
Sample Grade Card

SAMPLE FIGURES

Tomato • 200 kg lot

A

GRADE



- Size: uniform
- Colour: red, ripe
- Damage: under 5%
- Cleanliness: little dust

₹28-32 per kg vs ₹22 ungraded • FPO verified

Takeaway

Quality gets paid when it can be shown at a glance: a verified Grade A card turns a hidden advantage into a visible premium.

Risks and Measures

Risks: over-grading, photo fraud, buyers ignoring cards, grade drift. Measure: dispute rate, share of lots sold at grade, premium captured, spot-check match rate.

THE PROBLEM

Farmers Struggle to Tell What Is Wrong With the Crop

90%+ accuracy reported for 100+ classes in an Andhra Pradesh app study

17 apps reviewed in one study, with quality from excellent to unusable

Lab data popular training sets use plain-background photos; field accuracy can drop



Look-Alike Symptoms

Disease, pests, nutrient shortage and weather stress can look alike, so one photo can mislead.



Advice That Is Sold

The nearest adviser is often the input shop, and the advice may come with a product to sell.



Experts Are Far

Extension officers and KVK scientists cover wide areas, so answers can come late.



Overconfident Answers

Many apps give one confident diagnosis without saying how sure they are, and some carry ads.

Why Current Options Fall Short

AI diagnosis apps: Strong on common problems, but typically give one answer and some carry ads

Dealer advice: Fast and local, but tied to what the shop sells

Extension and Kisan Call Centre: Trusted and unbranded, but slower, and a phone call cannot see the leaf

THE SOLUTION, SIMPLIFIED

Crop Photo Check: Likely Issues, Honest Advice

1 • SEE

Better Photos, Fewer Misses

Problem: One blurry leaf photo cannot separate look-alike problems.

- Guided photos: leaf front, leaf back, whole plant
- Three quick questions: crop stage, spread, recent spray
- Telugu prompts with picture examples

Goal: enough context for a fair guess

2 • ANSWER

Top Three, Not One Verdict

Problem: A single confident answer misleads when symptoms look alike.

- Up to three likely causes, marked likely or possible
- Says "not sure" and routes low-confidence cases to an expert
- Checked against state university advisories

Goal: honest about uncertainty

3 • ADVICE

Unbranded, Cost-Aware Steps

Problem: Advice tied to a product pushes farmers to spend on what they may not need.

- Low-cost steps first: sanitation, water, nutrition checks
- No brand names; sprays only per state recommendations
- Free handoff to a KVK or Kisan Call Centre

Goal: advice a farmer can trust

Pilot One crop and about eight common issues, run on WhatsApp through an FPO. An agronomist reviews 1 in 10 replies. Add crops only as expert agreement holds up. Scope is illustrative.

HOW IT WORKS (ILLUSTRATIVE)

From a Leaf Photo to Honest Advice

Check Flow

1. **Send:** Farmer sends leaf, underside and whole-plant photos on WhatsApp
2. **Ask:** Bot asks three short questions about stage, spread and recent spray
3. **Match:** Model ranks likely causes and checks them against local advisories
4. **Reply:** Top three issues with confidence, cheap first steps, no brands
5. **Escalate:** If unsure, an agronomist or KVK takes over and the case is logged

Sample WhatsApp Reply

SAMPLE ONLY

Chilli leaf: likely issues

1. Leaf curl from sucking pests or virus (likely)

2. Nutrient shortage (possible)

3. Water or heat stress (possible)

Check first: underside of leaves for tiny insects, and soil moisture.

Do now: remove badly hit plants, keep weeds down, avoid over-watering. No spray yet.

Unsure? Send two more photos or call your KVK.

Guidance only, not a lab diagnosis. No brands recommended.

Takeaway

Farmers get a fair shortlist and cheap first steps, with honest uncertainty, instead of a confident guess or a product pitch.

Risks and Measures

Risks: misdiagnosis, overtrust, outdated advice, poor photos. Measure: expert agreement on the top three, correct "not sure" handoffs, spend avoided, repeat use.

THE PROBLEM

Organic Produce Rarely Reaches a Real Buyer

~5,000 FPOs registered on ONDC to sell online (2024 govt. statement)

50+ Rythu Bazaars listed on Digi Rythu Bazaar (platform figure)

5 km initial delivery radius in the Visakhapatnam pilot



Trust Gap

Buyers cannot easily verify "organic". Without visible proof, it sells like ordinary produce.



Middle Chain

Organic lots often merge into conventional supply chains, where traders and aggregators keep the premium.



Small, Perishable Lots

Scattered small volumes are hard to aggregate, grade and deliver fresh, and logistics eats the margin.



No Price Premium

Rythu Bazaar prices are government-decided. That suits regular produce but rewards no organic premium.

Why Current Options Fall Short

Digi Rythu Bazaar: Direct farm-to-home pilot at set prices, but no organic tier or premium

ONDC: Open network with FPOs onboarded, but buyers still need trust and reliable delivery

Certification: NPOP and PGS-India exist, but buyers rarely see proof at the point of sale

THE SOLUTION

An Organic Marketplace on ONDC and Digi Rythu Bazaar

1 • TRUST

Verified Organic Identity

Problem: Buyers cannot tell real organic from a label, so honest farmers earn no premium.

- Listings show NPOP or PGS-India certificate status
- QR code traces each lot to farm and harvest date
- FPO vouches for members and checks lots at the hub

Goal: buyers trust the label

2 • REACH

ONDC as the Open Storefront

Problem: Digi Rythu Bazaar serves a few zones, so most organic farmers have no route to distant buyers.

- Rythu Bazaar hubs act as seller nodes on ONDC
- Any ONDC buyer app can discover organic listings
- Local delivery from hubs; ONDC logistics beyond

Goal: any buyer app, any city

3 • FAIR PRICE

Direct Pricing, No Middle Chain

Problem: Fixed prices give no premium and traders capture the margin between farm and plate.

- Organic tier priced as a band above the Rythu Bazaar rate
- UPI payment settles direct to farmer or FPO
- Weekly subscription baskets give steady demand

Goal: the farmer keeps the premium

Pilot scope Start in Visakhapatnam, where Digi Rythu Bazaar already delivers. Onboard 20-30 certified organic farmers through FPOs, list them on ONDC, and fulfil from the Rythu Bazaar hub. Scope is illustrative.

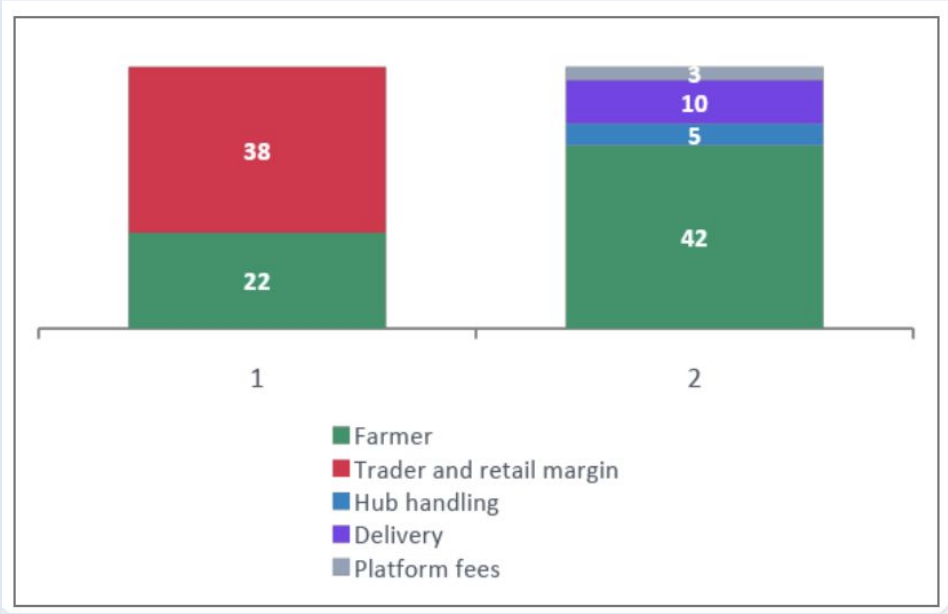
HOW IT WORKS (ILLUSTRATIVE)

From Organic Farm to Doorstep in Five Steps

Order Flow

1. **List:** Farmer or FPO lists certified lots with a QR trace on an ONDC seller app
2. **Discover:** Buyer finds them on Digi Rythu Bazaar or any ONDC buyer app
3. **Pack:** Order routes to the nearest Rythu Bazaar hub for weighing and packing
4. **Deliver:** Delivery partner brings it home; UPI pays the farmer or FPO directly
5. **Repeat:** Ratings and weekly baskets build trust and steady demand

Where ₹60 for 1 kg of Organic Tomatoes Goes (₹)



Takeaway

At the same ₹60 consumer price, the farmer share rises from ₹22 to ₹42 because a hub, delivery and small fees replace the trader and retail layers.

Risks and Measures

Risks: false organic claims, perishability and delivery cost, thin demand outside cities. Measure: farmer share of price, repeat-order rate, spoilage and premium versus conventional.

AUTONOMOUS AGRI-ROBOTICS INITIATIVE

Universal Robotic Farming Stack

Why Build KisaanBot?

Growing fresh produce and mastering farming should not be hindered by steep learning curves or daily manual friction. We are building KisaanBot to fundamentally minimize the friction to grow our food and learn farming—empowering growers and students through accessible, precision-guided automation.

What Are We Building?

An open-source, modular cyber-physical platform combining a Cartesian CNC motion base with interchangeable smart tool-heads, edge computer vision, and autonomous agronomic decision support.

Millimeter CNC
Sub-mm XY Repeatability

Standard Pogo Bus
Hot-Swappable Tools

Edge AI + Vision
YOLO & Telemetry DSS



REFERENCE & DEMONSTRATION

FarmBot Genesis: Technical Reference Breakdown



Key Video Highlights & Timestamps

0:50 – 1:30	**Hardware Architecture:** Track, gantry, and Cartesian 3-axis motion design overview.
1:31 – 2:15	**Universal Tool Mount:** Automatic tool-changing mechanism and magnetic pin bus demo.
2:16 – 3:00	**Precision Planting & Dosing:** Automated seed injection and targeted micro-watering.
3:01 – 3:45	**Computer Vision Integration:** Camera calibration and weed detection sequence.

Engineering Takeaways

The FarmBot Genesis architecture serves as a baseline proof-of-concept for Cartesian farm robotics. advances this concept by introducing sub-millimeter CNC repeatability, hot-swappable Pogo Bus tool-heads, and real-time edge AI telemetry to fully automate agricultural decision-making.

BUILDATHON ENGINEERING TRACKS

Three Core Subsystem Challenges

CHALLENGE 1 • ECE / MECH

Universal Tool-Head & Smart Pogo Bus

Problem: Swapping tools currently requires manual cable rewiring and fluid plumbing, risking short-circuits and leaks with zero digital verification of which tool is mounted.



- **Kinematic Mount:** 3-point magnetic auto-docking
- **EEPROM Auto-Discovery:** 1-Wire / I2C tool handshake
- **Leak-Free Dosing:** Auto-valved quick-disconnect line

Target: 20 auto-dock cycles & ≤±5% liquid variance

CHALLENGE 2 • CS / AI

FieldVision Homography & Scanner

Problem: Camera perspective tilt and lens warping cause standard pixel detections (u,v) to drift by 15–20 mm, causing tools to miss plant targets or damage seedlings.



- **ArUco Calibration:** 4-point bird's-eye warp matrix
- **Lightweight YOLO:** Crop centroid & weed detection
- **G-code Auto-Generation:** Stream direct G00 XY targets

Target: ≤2.5 mm spatial accuracy & <2s inference

CHALLENGE 3 • CS / SYSTEMS

Universal Agronomic DSS & Bridge

Problem: Field robots execute hardcoded timer routines blind to real soil hydration and weather, resulting in overwatering, wasted battery, and zigzag travel paths.



- **Weather Gating:** Delay watering on incoming rain
- **TSP Path Solver:** Shortest travel across 10+ crops
- **Dual Output:** Simultaneous GRBL & Nav2 compatibility

Target: 100% automated dispatch & zero manual telemetry entry